

IILM UNIVERSITY

SECURE MULTI-TENANT LEGAL TECHNOLOGY PLATFORM DEVELOPMENT

Student	AMRITESH GAUTAM
Enrollment No.	CS2341800
Program & Semester	B.Tech CSE - Semester VII
Organization	Clearpaper
Role	Independent Software Developer

Batch: 2023–27 | Section: 4CSE3 | Academic Year: 2026–27

1. Internship & Organization

Organization

Early-stage legal technology startup based in London, United Kingdom, developing a SaaS platform for legal workflows.

Engagement

Independent Software Developer
Remote • Part-time
From 1 June, 2026
Till 1 August, 2026

Development Context

Small development team working on an actively evolving product with emphasis on security, reliability, maintainability and rapid iteration.

Focus of the internship

- Contributing to a secure legal-technology SaaS platform.
- Working across backend, frontend, data, document infrastructure and integrations.
- Supporting selected AI-assisted workflows while keeping unreleased product details confidential.

2. Problem Statement & Objectives

Problem Statement

Legal professionals handle large volumes of sensitive documents and matter-related information. Conventional file-sharing may not provide sufficient structure around matters, roles, permissions, versions and auditability.

Project Objectives

- Secure multi-tenant SaaS development
- Backend services, APIs and data management
- Authenticated, matter-based frontend workflows
- Secure document storage and version management
- Permission-aware application behaviour
- Integrations and workflow automation
- Selected AI-assisted activity and time-tracking workflows
- Reliability, usability and maintainability

3. Scope, Technologies & Tools

Project Scope

Full-stack development across backend APIs, database-backed logic, frontend interfaces, authentication and authorization, document management, integrations and selected automation/AI-assisted workflows.

Engineering Responsibilities

Cloud-backed storage, document-processing pipelines, permission-aware operations, data consistency, iterative development and adaptation to changing requirements.

TypeScript / Node.js
Backend

Express
REST APIs

Next.js / React
Frontend

PostgreSQL / Prisma
Data

AWS S3
Object storage

Microsoft Graph
Integrations

PDF / Text Libraries
Document processing

Git / GitHub
Source control

4. High-Level Architecture & Methodology

Web Application
User interface

Backend API
Auth, validation & business logic

Database
Structured application data

Object Storage
Secure document files

Document Processing
Text extraction

External Integrations
Selected external services

Iterative methodology

1
Requirement understanding

2
Design

3
Implementation

4
Integration

5
Verification

6
Iteration

5. Outcomes & Key Takeaways

- Practical understanding of production-oriented full-stack software development.
- Hands-on experience with secure multi-tenant workflows, authorization and document management.
- Development experience with TypeScript, Node.js, Express, Next.js, React, PostgreSQL and Prisma.
- Exposure to cloud object storage, document processing and external API integrations.
- Experience contributing to AI-assisted workflows in an evolving product environment.
- Improved ability to work independently, communicate technical decisions and deliver features within a small team.

Thank You